



The role of artificial intelligence in enhancing firm investment efficiency

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ABSTRACT

This paper investigates the effect of artificial intelligence (AI) on firm investment efficiency. The results show that AI can significantly enhance the investment efficiency of enterprises. This effect is more pronounced in high knowledge-intensity firms, high R&D-intensity firms, and firms with sound internal controls. Mechanism analysis reveals that AI contributes to improved investment efficiency by enhancing innovation capabilities and strengthening the financial health of firms. Furthermore, we find that AI adoption may lead to a rise in firm value. This study adds new insights to how AI influences firms' financial and strategic decision-making, providing a valuable reference for policymakers. It offers guidance for the government in formulating policies that promote intelligent transformation and encourage enterprises to leverage AI for innovation and improved investment efficiency.

1. Introduction

Artificial intelligence (AI) is one of the most significant achievements of 21st-century science and technology and a key driver of progress in the information age. Today, AI plays a leading role in industrial transformation, technological innovation, and economic modernization, profoundly impacting the economy, society, and everyday life. By the end of 2024, China had more than 4500 AI companies, with the core AI industry valued at nearly 600 billion RMB. Moreover, China's authorities have released multiple strategic guidelines, the "Smart Manufacturing Development Plan (2016–2020)", emphasizing the importance of intelligent production technologies in driving the manufacturing industry's shift toward high-quality growth.

In recent years, investment activity among Chinese manufacturing firms has steadily increased; however, enhancing investment efficiency continues to pose major challenges. Notable regional imbalances persist, and a significant number of enterprises operate below optimal efficiency levels. As key drivers of the real economy, these firms must improve

their investment efficiency—not only to ensure their own long-term viability but also to support the broader objective of fostering high-quality national economic growth. Prior research has mainly focused on issues such as weak internal controls, information asymmetry, limited access to financing, and agency conflicts as determinants of investment inefficiency (Denis & Sibilkov, 2010; Liu & Yu, 2023; McLean, Zhang, & Zhao, 2012; Stein, 2003; Zheng & Xiao, 2020). While artificial intelligence (AI) has attracted increasing attention in corporate strategy and performance discussions, its specific impact on investment efficiency remains insufficiently examined. This study seeks to explore how the adoption of AI technologies influences the efficiency of corporate investment decisions. Using panel data from Chinese A-share listed companies between 2008 and 2023, we also analyze the mechanisms through which AI may alter firms' investment behavior.¹

This paper argues that artificial intelligence enhances corporate investment efficiency through two primary mechanisms. First, AI boosts investment efficiency by strengthening firms' innovation capabilities. One of AI's most significant advantages is its ability to extract deep

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¹ The Chinese context offers a unique and highly relevant setting for examining the impact of AI adoption on firm-level investment efficiency due to its distinct institutional, economic, and policy characteristics. Unlike many Western economies, China has pursued a top-down, state-led approach to technological development, with national initiatives such as "Made in China 2025" and the "Smart Manufacturing Development Plan (2016–2020)" serving as strong policy signals that actively shape corporate behavior. Additionally, China's corporate landscape features a large number of firms, significant regional disparities, and higher levels of information asymmetry—particularly among SMEs—resulting in more severe investment inefficiencies. In this environment, AI's potential to reduce frictions and enhance decision-making is likely to manifest differently than in more mature and transparent markets, making China an ideal context for this study.

insights from data, allowing companies to transcend traditional innovation boundaries. Increasingly, firms are integrating AI technologies across various stages of research and development (R&D), including the intelligent analysis of technical literature, the prediction of R&D project success rates, and virtual testing using digital twins. This is particularly valuable for innovation activities that involve high technological complexity and uncertainty. By improving the likelihood of innovation success and reducing the costs of trial and error in R&D, AI enables firms to better anticipate market demand and technological trends. This, in turn, facilitates more informed investment decisions, allowing firms to allocate limited resources to the most promising innovation areas, ultimately leading to a systematic improvement in investment efficiency.

Second, artificial intelligence (AI) enhances investment efficiency by improving a firm's financial performance. Leveraging big data analytics and intelligent algorithms, AI enables real-time monitoring and forecasting of financial conditions, helping managers overcome the limitations of traditional financial analysis and make more precise investment decisions. AI offers three key advantages in financial optimization: it can comprehensively integrate multidimensional financial data, conduct dynamic risk assessments with early warnings, and autonomously adjust financial strategies in response to market fluctuations. As a result, the use of AI in financial management enhances a firm's ability to process complex financial information, increases the precision of capital allocation, and boosts investment returns. Moreover, as AI technologies continue to evolve and become more deeply embedded in business operations, collaboration between AI systems and financial professionals is expected to yield new models of financial analysis. These innovative approaches will allow firms to assess investment projects more scientifically and mitigate financial risks, leading to a systematic improvement in investment efficiency.

The data sources for the main variables used in this study are as follows: The corporate investment efficiency is calculated using data on listed companies from the CSMAR database. Following the investment model proposed by Richardson (2006), we estimate the residuals from the model and take the negative absolute value of these residuals to measure investment efficiency—where smaller deviations from expected investment indicate higher efficiency. The corporate artificial intelligence (AI) adoption is measured through text analysis of company annual reports. First, we collect firms' annual reports from 2008 to 2023 from CNINFO (China Securities Market Financial Disclosure Network) by python web-scraping tools. Drawing on methodologies from Chen and Srinivasan (2020) and Li, Mai, Shen, and Yan (2021), we constructed an AI-related keyword dictionary based on resources from the World Intellectual Property Organization (WIPO). This dictionary includes 73 terms such as "artificial intelligence," "AI products," and "AI chips." Using Python's open-source word segmentation package Jieba, we performed text matching on the collected annual report data. The frequency of AI-related keywords appearing in each report was then counted, with the final measure of AI usage calculated as the natural logarithm of one plus the keyword frequency. After excluding ST and ST* firms, as well as observations with missing or incomplete data, the final dataset includes 24,768 firm-year observations.

However, the empirical results of this study may be subject to potential endogeneity issues. It is also important to consider the potential for reverse causality, as enterprises demonstrating greater investment efficiency might exhibit a higher propensity to implement AI technologies, thereby complicating the causal inference between AI adoption and investment performance. Although we have controlled for year and industry fixed effects and included a rich set of control variables, the results may still be influenced by omitted observable or unobservable factors, thus raising endogeneity concerns. To address these potential issues, we employ two strategies: First, we use the issuance of the "Smart Manufacturing Development Plan (2016–2020)" as an exogenous policy shock and conduct DID regressions to identify the impact of AI adoption on investment efficiency. Second, we construct a Bartik-style instrumental variable based on the shift-share method to further address

endogeneity concerns and perform two-stage regressions. The results from both methods support the validity of the baseline model, confirming the robustness of our main finding: artificial intelligence significantly enhances corporate investment efficiency. The results of the heterogeneity analysis indicate that the enhancement in investment efficiency associated with artificial intelligence is particularly evident among firms characterized by higher knowledge intensity, greater R&D commitment, and more robust internal control systems. Furthermore, the mechanism analysis shows that AI enhances investment efficiency primarily by boosting firms' innovation capabilities and improving their financial conditions.

This study makes several contributions to the existing literature. First, it enriches research on the impact of artificial intelligence on corporate economic behavior. While previous studies have mainly examined AI's influence on financial decision-making, operations management, risk control, and financial performance (Lee, Yang, & Kim, 2019; Li, Xu, Zheng, et al., 2023; Mishra, Chen, et al., 2022; Bhatia, 2019; Liu, Dong, Mei, & Shen, 2023), this paper extends the discussion by focusing specifically on how AI enhances corporate investment efficiency. In doing so, it contributes to the broader fields of digital finance and financial technology (FinTech). Furthermore, by constructing a tailored AI-related keyword dictionary and employing a novel text analysis approach, this study not only introduces an innovative method for measuring corporate AI adoption but also expands the theoretical understanding of AI's role in firm-level economic activity.

Second, while prior studies on the determinants of corporate investment efficiency have primarily focused on factors such as internal control deficiencies, information asymmetry, financing constraints, and agency problems (Denis & Sibilkov, 2010; Liu & Yu, 2023; McLean et al., 2012; Stein, 2003; Zheng & Xiao, 2020), this study shifts the focus to artificial intelligence—a transformative force at the forefront of the current technological and industrial revolution. By empirically demonstrating the role of AI in improving investment efficiency, this paper offers robust evidence of AI's productivity-enhancing potential at the firm level. In doing so, it contributes to the literature on corporate investment efficiency by introducing a novel, technology-oriented perspective.

The remainder of this paper is organized as follows: Section 2 provides a review of the existing literature and formulates the study's hypotheses. Section 3 outlines the data collection process and empirical methodology. Section 4 reports the core findings. Section 5 investigates the heterogeneity. Section 6 delves into the mechanisms. Section 7 discusses the broader economic consequence. Section 8 concludes.

2. Literature review and research hypotheses

2.1. Literature on AI's impact on corporate financial behavior and performance

As the core of the contemporary technological revolution, artificial intelligence (AI) is profoundly influencing corporate financial behavior and performance. AI, a branch of computer science that simulates human intelligence, refers to technologies and methods that employ computer programs, machines, and other means—such as machine learning, deep learning, and natural language processing—to emulate, extend, and expand human intelligence. AI can rapidly integrate and process vast amounts of data, providing technical support for the practice and innovation of corporate financial management. In recent years, a substantial body of literature has emerged on AI's applications in corporate financial behavior and performance, with research primarily focusing on financial decision-making, operational management, risk management, and financial performance.

In terms of corporate financial decision-making, AI alleviates financing constraints faced by firms and enhances the rationality of asset pricing, thereby improving the efficiency of investment decisions. The development of the internet has enabled corporate activities such as

production, sales, and payments to leave “digital footprints” online. Confronted with this complex data, banks can leverage AI to quickly identify and process credit-related information, accurately constructing a “digital profile” of firms. This improves the accuracy and timeliness of information processing (Gambacorta, Huang, Qiu, & Wang, 2019; Liberti & Petersen, 2019), thereby mitigating financing constraints caused by insufficient credit information for small and medium-sized enterprises (SMEs). Lee et al. (2019) also find that fintech technologies, represented by AI, significantly reduce financing gaps and optimize credit allocation. Additionally, AI can improve firms’ financing environments (Goldstein et al., 2019), help identify and evaluate investment opportunities, and enhance capital allocation efficiency (Vial, 2019; Acemoglu and Restrepo, 2019).

In the context of operational management, Li et al. (2023) find that AI significantly enhances resource allocation efficiency in their study of how resource reallocation impacts innovation efficiency among Chinese listed companies. Similarly, Mishra et al. (2022), analyzing reports from U.S. public firms, demonstrate that AI improves operational efficiency and return on investment while lowering marketing costs. Bharadwaj, Sawy, Pavlou, and Venkatraman (2013) argue that digital technological innovations—exemplified by AI—empower organizational management and help reduce internal control costs.

In risk management, numerous scholars have employed AI to predict corporate risks. For example, risk perception models and dynamic financial risk early-warning models have been constructed using natural language materials and technologies, integrating convolutional neural networks and long short-term memory networks (Bhatia, 2019). Ghodselahi and Amirmadhi (2011) design a hybrid credit scoring model applying ensemble learning to credit decisions, validating AI’s feasibility in credit risk management. Hasan, Taylor, and Richardson (2022) use gradient boosting tree algorithms to improve the accuracy of predicting future risks while studying the impact of brand capital on stock price crash risk in listed companies.

Regarding financial performance, digital technological innovations represented by AI can reduce information asymmetry among corporate divisions, thereby improving financial performance (Liu et al., 2023). Vrontis, Christofi, Pereira, et al. (2022) explore the impact of digital intelligence on human resource management and corporate performance, noting that AI and other intelligent technologies enhance resource management efficiency and reduce costs, leading to overall financial performance growth. Since humans and AI each have unique advantages, corporate management can coordinate their complementary strengths to improve performance (Davenport & Kirby, 2016). Additionally, many scholars have studied AI’s impact on total factor productivity, labor demand, new quality productivity, and corporate innovation. For instance, Babina, Fedyk, He, and Hodson (2024) propose a metric using employee resumes to measure AI workforce size at the company level, serving as a new indicator of AI investment. Using this metric, they found that AI promotes product innovation and drives corporate growth.

2.2. Literature on factors influencing corporate investment efficiency

Corporate investment efficiency refers to the gap between actual and ideal investment levels. A smaller gap indicates higher efficiency, while a larger gap reflects inefficiency, with the difference representing non-efficient investment. Existing research on factors influencing investment efficiency primarily focuses on internal controls, information asymmetry, financing constraints, principal-agent problems, ownership structure, and external environmental and policy issues (Zheng et al., 2023; Wang et al., 2024).

In terms of internal controls, Liu and Yu (2023) studied A-share listed companies in China, finding that internal control quality significantly reduces non-efficient investment. Deumes and Knechel (2008) argued that improving internal control quality enhances external confidence in accounting information, reducing financing costs and non-efficient

investment.

Regarding information asymmetry, Stein (2003) systematically explained its impact on investment decisions, noting that it disrupts capital allocation efficiency. According to Biddle and Hilary (2006), enhanced accounting quality plays a role in mitigating information asymmetry, thereby facilitating more efficient investment decisions. Similarly, Chen, Hope, Li, and Wang (2011) provide evidence of a positive association between the quality of financial reporting and firms’ investment efficiency. McNichols and Stubben (2008) showed that earnings management distorts accounting information, leading to inefficient investment decisions. Gomariz and Ballesta (2014) studied Spanish listed companies (1998–2008), concluding that financial reporting quality alleviates overinvestment. Other studies found that digital finance, ESG, and key disclosure items also reduce information asymmetry, enhancing investment efficiency.

On financing constraints, McLean et al. (2012) found that firms facing higher constraints exhibit greater sensitivity of investment to internal cash flows but weaker responsiveness to investment opportunities. Market imperfections exacerbate financing constraints, leading to inefficient investment (Campello, Graham, & Harvey, 2010). Local government debt expansion crowds out access to quality bank loans, causing firms to miss investment opportunities (Cao, Duan, & Ibrahim, 2023; Huang, Pagano, & Panizza, 2020; Ma & Hou, 2023). Regarding principal-agent problems, Zheng and Xiao (2020) used agency theory to show that managerial corruption suppresses investment. Denis and Sibilkov (2010) argued that financing constraints can reduce agency costs and curb overinvestment. Due to misaligned interests, agents may overinvest to build “empires,” seeking greater control benefits and professional reputation (Richardson, 2006). In ownership structure, Chen, Ghoul, Guedhami, and Wang (2017) examined how state and foreign ownership affect investment efficiency in privatized firms globally. On external factors, Li, Li, and Wu (2019) found that exchange rate uncertainty stimulates investment in labor-intensive and low irreversibility firms using Chinese industrial data. Chen, Sun, Tang, and Wu (2011) showed that government intervention exacerbates inefficient investment in Chinese listed firms. Dai, Hou, and Li (2021) argued that policy incentives may induce irrational overinvestment.

Our paper investigates the influence of AI on corporate investment efficiency. Specifically, it constructs an indicator of corporate AI adoption using the word frequency method, measures corporate investment efficiency levels based on Richardson’s (2006) actual investment level model, and conducts empirical analysis. Secondly, the “Smart Manufacturing Development Plan (2016–2020)” policy enactment is used as an exogenous event shock in the DID method, along with a Bartik instrumental variable constructed via the shift-share method for endogeneity testing. Furthermore, robustness tests are performed by replacing the explained and explanatory variables. Additionally, this paper conducts heterogeneity analyses based on corporate location, firm type, and firm size, distinguishing between over-investment and under-investment scenarios for categorical testing of corporate investment efficiency. Finally, the paper explores transmission mechanisms through corporate innovation levels and financial conditions, followed by economic consequence testing. Current research on AI primarily focuses on the macro level, and due to the lack of effective firm-level measurement indicators for AI technology, how AI affects the investment efficiency of micro-level firms remains inconclusive. This paper builds upon existing research by constructing a new indicator for measuring corporate AI adoption, thereby expanding the literature on AI’s effect on firm economic behavior and enriching research on the determinants of firm investment efficiency.

2.3. Hypotheses development

Artificial intelligence refers to a class of technologies and methods that simulate, extend, and expand human intelligence. Artificial intelligence can influence corporate investment efficiency by improving

firms' financing environment (Goldstein et al., 2019), enhancing innovation capabilities (Babina et al., 2024), and optimizing financial conditions (Liu et al., 2023), among other pathways.

First, regarding corporate innovation capability. Issues such as lengthy R&D cycles, high trial-and-error costs, and uncertain technological pathways often constrain the efficient allocation of innovation resources, leading to low investment efficiency. Artificial intelligence can optimize R&D direction through big data analysis, predict technology trends, and reduce ineffective investments. Meanwhile, machine learning algorithms can accelerate experimental simulations, shorten innovation cycles, and lower trial-and-error costs. Additionally, AI-driven intelligent decision-making systems can optimize R&D resource allocation, thereby improve the conversion rate of innovation outcomes and enhancing corporate investment efficiency.

Second, in terms of corporate financial management. AI with big data analytics and machine learning technologies at its core, breaks through the temporal and spatial limitations of traditional financial management, thereby improving firms' data collaboration and intelligent decision-making capabilities. After obtaining and analyzing operational data, industry trends, and financial market information in real-time through AI systems, firms can more accurately assess and forecast their financial conditions, leading to more scientific investment decisions and higher resource allocation efficiency. In summary, our paper proposes the following hypothesis:

Hypothesis H1. Artificial intelligence can improve firm investment efficiency.

3. Research design

3.1. Data sources

This paper selects A-share listed companies from the Shanghai Stock Exchange and Shenzhen Stock Exchange from 2008 to 2023 as sample data. The following treatments are applied to the sample data: First, ST and ST* listed companies are excluded; second, financial-type listed companies are removed; third, observations with missing financial data are eliminated. The annual report data of enterprises used in this study are sourced from the CSMAR database and the China Securities Market Financial Disclosure Network (CNINFO); patent data are obtained from the International Intellectual Property Rights Database (IRPDB); basic corporate information and financial data come from the CSMAR database; GDP data are sourced from the National Bureau of Statistics; regional marketization level data are extracted from the China Marketization Index Database. Furthermore, all continuous variables are subjected to 1 % winsorization to mitigate the influence of outliers. Following this data preprocessing, the final dataset comprises 24,768 firm-year observations.

3.2. Variable selection and descriptive statistics

3.2.1. Dependent variable

The dependent variable in this paper is the opposite number of corporate inefficient investment level, which serves as a proxy for investment efficiency (*Inveff*). Following Richardson's (2006) actual investment level model,² the residual is calculated by constructing the following model to measure corporate inefficient investment level:

$$Invest_{i,j,t} = \alpha_0 + \alpha_1 Invest_{i,j,t-1} + \alpha_2 Controls_{i,j,t-1} + \sum Industry + \sum Year$$

² Referring to Richardson (2006) in the paper "Over-investment of Free Cash Flow," this study selects Invest and the lagged Invest as the explained variable and explanatory variable, respectively, measured by the firm's new capital expenditures. *Leverage*, *Growth*, *Size*, *Age*, *Cash*, and *Return* are chosen as control variables, measured by the asset-liability ratio, operating income growth rate, natural logarithm of total assets, firm age, cash holding ratio, and stock excess return rate, respectively, to estimate the residual term.

+ $\varepsilon_{i,j,t}$ (1).

Where: Subscripts i, j, and t represent individual firms, industries, and years, respectively. Controls include *Leverage*, *Growth*, *Size*, *Age*, *Cash*, and *Return*. The path dependence effect of historical investment scale directly influences current resource allocation decisions, while financial leverage affects investment behavior through dual channels of financing constraints and agency costs. Growth opportunities and firm size respectively reflect future demand potential and resource allocation capabilities, with firm age capturing risk preference differences arising from the life cycle. Cash holdings reveal the liquidity constraints on investment flexibility, and stock returns internalize the shaping effect of market signals on managerial expectations. Additionally, by introducing industry fixed effects and year fixed effects, we control for systematic interference from industry heterogeneity and macroeconomic shocks on investment behavior. All variables adopt a one-period lag design to mitigate endogeneity issues between contemporaneous variables and investment decisions. The residuals ($\varepsilon_{i,j,t}$) obtained from the fitted model represent the difference between a firm's actual investment level and its expected investment level, serving as a measure of inefficient investment. This study employs the negative value of the absolute residual to gauge investment efficiency (*Inveff*), where a higher value indicates greater investment efficiency and a lower value signifies higher inefficiency. A residual with a positive value reflects an overinvestment scenario, whereas a negative residual suggests insufficient investment relative to expectations.

3.2.2. Explanatory variables

The explanatory variable in this paper is the status of corporate artificial intelligence (AI) usage (AI). Whether a company develops AI depends on management decisions, and the annual reports of listed companies reflect the importance they place on AI. Generally, the higher the frequency of AI-related terms in a listed company's annual report, the greater the emphasis the company places on AI development. Therefore, the AI-related vocabulary disclosed in annual reports can, to some extent, represent the status of corporate AI usage. Drawing on the AI dictionary construction methods of Chen and Srinivasan (2020) and Li et al. (2021), to quantify corporate engagement with AI, this study identifies a set of AI-related keywords—such as artificial intelligence, AI products, AI chips, machine translation, machine learning, computer vision, human-computer interaction, deep learning, neural networks, biometrics, and image recognition—and matches them with textual content from firms' annual reports. The natural logarithm of one plus the frequency of these terms (*lnword*) serves as the primary proxy for AI adoption. For robustness checks, the same set of AI-related terms is cross-referenced with firms' patent data to determine the number of relevant AI patents granted. The natural logarithm of one plus the patent count (*lnpatent*) is used as an alternative indicator of AI involvement.

Specifically, the construction process of the explanatory variables in this paper is as follows:

I. Data Collection.

Python web scraping technology is employed to collect annual reports of listed companies from the China Securities Market Financial Disclosure Network (CNINFO) for the years 2008–2023, which are then converted into txt-format data. Similarly, the API interface provided by the International Intellectual Property Rights Database (IRPDB) is used to scrape the titles and abstracts of patents held by listed companies from 2008 to 2023, which are also converted into txt-format data.

II. Construction of the AI Dictionary.

The AI vocabulary selected in this paper is based on the research of Chen and Srinivasan (2020) and Li et al. (2021). The construction of the AI-related term dictionary follows a three-step procedure: (1) An initial list of 52 seed terms is compiled by drawing on sources such as the World Intellectual Property Organization's AI vocabulary and industry analyses, including the 'STAR Market Series—Panoramic View of the AI Industry Chain' by Ping An Securities. (2) Using the Skip-gram model, word embeddings are trained on corpora composed of annual reports

and patent documents. For each seed term, the ten most semantically similar words are identified based on cosine similarity. (3) After eliminating duplicates, contextually irrelevant terms, and low-frequency entries, a refined set of 73 AI-related terms is obtained as the final dictionary.

III. Generation of AI Variables.

The finalized AI dictionary is incorporated into the open-source Python package ‘jieba’ for word segmentation. Subsequently, the frequency of AI-related terms within annual reports is calculated, and the natural logarithm of one plus this frequency serves as the primary metric for measuring corporate AI adoption. In parallel, AI-related patents are identified and used as an alternative measure to verify the robustness of the results.

3.2.3. Control variables

With reference to existing literature on the influencing factors of corporate investment efficiency, this paper sets corporate-level control variables. Specifically, the control variables include: listing age, asset-liability ratio, return on assets, Tobin’s Q, firm size, the proportion of shares held by the largest shareholder, CEO duality, net cash flow from operating activities, board size, audit opinion, and the proportion of independent directors. Additionally, this paper controls for fixed effects at the industry and year levels.

3.2.4. Descriptive statistics

Table 1 summarizes the descriptive statistics. The average corporate investment efficiency (*Inveff*) stands at -0.039 with a standard deviation of 0.040 , indicating a widespread presence of investment inefficiency among firms in the sample period and notable variation in investment efficiency across companies. Firm AI adoption, measured by *lnword*, has a mean value of 0.293 and a standard deviation of 0.647 , reflecting a generally low level of AI utilization on average, alongside substantial heterogeneity in AI engagement among firms. Other variables included in the analysis exhibit descriptive statistics within expected and reasonable bounds.

3.3. Model construction

This paper employs a fixed-effects model to examine the impact of artificial intelligence on corporate investment efficiency. The regression model is specified as follows:

$$Inveff_{i,j,t} = \alpha_0 + \alpha_1 lnword_{i,j,t-1} + \alpha_2 X_{i,j,t} + \lambda_j + \mu_t + \varepsilon_{i,j,t} \quad (2).$$

In Model (2), the subscripts i , j , and t represent individual firms, industries, and years, respectively. Here, $Inveff_{i,j,t}$ denotes the investment efficiency of firm i , while $lnword_{i,j,t-1}$ indicates the artificial intelligence usage status of firm i . To mitigate endogeneity issues arising from reverse causality, this study applies a one-period lag to the variable measuring firms’ artificial intelligence usage. $X_{i,j,t}$ represents firm-level

control variables, which specifically include: firm age, capital structure, return on assets, Tobin’s Q, firm size, the shareholding ratio of the largest shareholder, CEO duality, net cash flow from operating activities, board size, audit opinion, and the proportion of independent directors. λ_j and μ_t control for industry and year fixed effects, respectively, and $\varepsilon_{i,j,t}$ is the random error term.

4. Empirical analysis

4.1. Analysis of empirical results

This study performs regression analysis on the sample data, with the baseline results displayed in Table 2. Columns (1) through (4) report firm-clustered regression estimates under varying specifications: column

Table 2
Impact of Artificial Intelligence on Corporate Investment Efficiency.

Variable	(1) <i>Inveff</i>	(2) <i>Inveff</i>	(3) <i>Inveff</i>	(4) <i>Inveff</i>
<i>lnword</i>	0.004*** (0.001)	0.005*** (0.001)	0.003*** (0.001)	0.003*** (0.001)
<i>Age</i>		0.004*** (0.001)	−0.002** (0.023)	−0.001* (0.081)
<i>Leverage</i>		−0.022*** (0.003)	−0.012*** (0.005)	−0.015*** (0.006)
<i>ROA</i>		−0.090*** (0.007)	−0.082*** (0.008)	−0.085*** (0.008)
<i>TQ</i>		−0.002*** (0.001)	−0.001*** (0.001)	−0.001*** (0.001)
<i>Size</i>		0.004*** (0.001)	0.003*** (0.001)	0.003*** (0.001)
<i>Top</i>		−0.003 (0.223)	−0.002 (0.301)	−0.002 (0.286)
<i>Dual</i>		0.001 (0.659)	0.002** (0.032)	0.002** (0.034)
<i>NCF</i>		0.019*** (0.003)	0.012*** (0.005)	0.018*** (0.004)
<i>Director</i>		−0.002 (0.321)	−0.003* (0.091)	−0.002 (0.227)
<i>Audit</i>		0.001 (0.576)	0.001 (0.598)	0.001 (0.602)
<i>RID</i>		0.008* (0.084)	0.007 (0.234)	0.006 (0.304)
<i>Constant</i>	Yes	Yes	Yes	Yes
<i>Industry FE</i>	Yes	Yes	No	Yes
<i>Year FE</i>	Yes	No	Yes	Yes
<i>R²</i>	0.105	0.102	0.044	0.113
<i>Observations</i>	24,768	24,768	24,768	24,768

This table shows the regression results of the effect of AI on corporate investment efficiency, with the explained variable being *Inveff*. ***, **, and * denote significance at the 1 %, 5 %, and 10 % levels, respectively.

Table 1
Descriptive Statistics.

Variable	Sample size	Mean	Standard deviation	Minimum value	Median	Maximum value
<i>Inveff</i>	24,768	−0.039	0.040	−0.247	−0.029	0
<i>lnword</i>	24,768	0.293	0.647	0	0	3.091
<i>Age</i>	24,768	2.416	0.811	0	2.639	3.401
<i>Leverage</i>	24,768	0.453	0.201	0.059	0.455	0.879
<i>ROA</i>	24,768	0.041	0.055	−0.151	0.036	0.213
<i>TQ</i>	24,768	1.892	1.122	0.826	1.534	7.102
<i>Size</i>	24,768	22.510	1.352	19.953	22.340	26.398
<i>Top</i>	24,768	0.346	0.150	0.085	0.326	0.740
<i>Dual</i>	24,768	0.797	0.402	0	1	1
<i>NCF</i>	24,768	0.051	0.066	−0.143	0.048	0.244
<i>Director</i>	24,768	2.307	0.238	1.609	2.303	2.890
<i>Audit</i>	24,768	0.979	0.145	0	1	1
<i>RID</i>	24,768	0.376	0.068	0.250	0.364	0.583

This table presents the descriptive statistics of sample enterprises during the period 2006–2014, including sample size, mean, standard deviation, minimum value, median, and maximum value.

(1) excludes control variables but includes industry and year fixed effects; column (2) incorporates control variables with industry fixed effects only; column (3) includes control variables with year fixed effects only; and column (4) controls for both industry and year fixed effects while including control variables.

The regression findings consistently demonstrate a statistically significant positive relationship between artificial intelligence adoption and corporate investment efficiency at the 1 % significance level, regardless of whether industry fixed effects, year fixed effects, or both are controlled for. This suggests that AI implementation contributes to improvements in firms' investment efficiency. From an economic perspective, after accounting for other variables and controlling for both industry and year effects, a one-unit increase in AI usage corresponds to an average enhancement of approximately 0.3 % in investment efficiency. Additionally, the estimated coefficients of the control variables align well with theoretical expectations.

4.2. Endogeneity test

4.2.1. Difference-in-differences method

This section selects the promulgation of the “Smart Manufacturing Development Plan (2016–2020)” policy as an event shock. The policy document emphasizes the significant role of intelligent manufacturing in the high-quality development of the manufacturing sector at the macro level. However, macro-level intelligent manufacturing planning must also be implemented at the micro-level of enterprises. Under the guidance of this policy, the Ministry of Industry and Information Technology of the People's Republic of China announced intelligent manufacturing demonstration projects for four consecutive years from 2015 to 2018. This paper treats enterprises included in the intelligent manufacturing demonstration projects as the treatment group and other enterprises as the control group, employing the staggered difference-in-differences method for robustness testing. The model specification is as follows:

$$Inveff_{i,j,t} = \alpha_0 + \alpha_1 Treat_{i,j} \times Post_t + \alpha_2 X_{i,j,t} + \lambda_j + \mu_t + \varepsilon_{i,j,t} \quad (3).$$

In Model (3), the subscripts i , j , and t represent the individual enterprise, the industry to which the enterprise belongs, and the year, respectively. When an enterprise is listed in the smart manufacturing demonstration project roster, $Treat_{i,j}$ is defined as the treatment group and assigned a value of 1; otherwise, it is the control group with a value of 0. Similarly, samples before and after the policy enactment are assigned $Post_t$ values of 0 and 1, respectively. The definitions of $Inveff_{i,j,t}$, $X_{i,j,t}$, λ_j , μ_t , and $\varepsilon_{i,j,t}$ are the same as in Model (2). The regression results are shown in Table 3.

The regression results show that the coefficient of $Treat \times Post$ on corporate AI adoption is significantly positive, indicating that the promulgation of the policy document “Smart Manufacturing Development Plan (2016–2020)” promotes improvement in corporate investment efficiency. The regression results corroborate our baseline regression results. To ensure the reliability of the DID outcomes, this study additionally performs a parallel trends assessment, presented in Table 4.

Table 3
Results of the Difference-in-Differences Test.

Variable	<i>Inveff</i>
<i>Treat</i> × <i>Post</i>	0.005*** (0.002)
Control variable	Yes
Constant	Yes
Industry FE	Yes
Year FE	Yes
R ²	0.114
Observations	24,028

This table shows the DID results on the impact of artificial intelligence on corporate investment efficiency, with the explained variable being *Inveff*.

Table 4
Parallel Trend Test Results.

Variable	<i>Inveff</i>
<i>Treat</i> × <i>Post</i> _{−3}	−0.001 (0.323)
<i>Treat</i> × <i>Post</i> _{−2}	−0.005 (0.265)
<i>Treat</i> × <i>Post</i> ₀	0.003 (0.175)
<i>Treat</i> × <i>Post</i> ₁	0.010*** (0.003)
<i>Treat</i> × <i>Post</i> ₂	0.007** (0.013)
<i>Treat</i> × <i>Post</i> ₃	0.005** (0.012)
Control variable	Yes
Constant	Yes
Industry FE	Yes
Year FE	Yes
R ²	0.114
Observations	24,028

This table presents the parallel trend test results in the DID method, with the explained variable being *Inveff*.

The above regression results align with the parallel trend assumption.

Meanwhile, to rule out the possibility that the improvement effect of intelligent manufacturing on corporate investment efficiency might be due to a placebo effect caused by other random factors unrelated to the policy, this paper assumes a unified policy enactment year of 2017 to conduct a placebo test. The results of the placebo test are shown in Table 5.

The results of the placebo test indicate that the $Treat \times Post$ term, with 2017 selected as the policy enactment year, is not statistically significant, passing the placebo test. This suggests that the enhancement in corporate investment efficiency reflects the actual impact of intelligent manufacturing implementation rather than being driven by other random factors.

4.2.2. Instrumental variable approach

To address potential endogeneity concerns stemming from measurement errors, omitted variables, and reverse causality, this study adopts the Bartik instrumental variable approach, following the methodology proposed by Blanchard and Katz (1992). The Bartik instrument leverages the shift-share technique, using aggregate growth rates combined with initial shares of variable units to generate predicted values. These predicted values are strongly correlated with the actual variable but remain exogenous by being uncorrelated with the error term, thereby providing a valid instrument for endogeneity testing. First, the annual AI-related term frequencies of firms are aggregated to compute the national AI growth rate for each year. Second, the national AI growth rate plus one is multiplied by the firm's lagged AI adoption status. Specifically, the construction formula for the Bartik instrumental variable is as follows:

Table 5
Placebo Test Results.

Variable	<i>Inveff</i>
<i>Treat</i> × <i>Post</i>	0.003 (0.252)
Control variable	Yes
Constant	Yes
Industry FE	Yes
Year FE	Yes
R ²	0.114
Observations	24,028

This table presents the placebo test results, with the explained variable being *Inveff*.

$$IVlnword_{i,j,t} = lnword_{i,j,t-1} \times (1 + lnwordGrowth_t) \quad (4).$$

In Model (4), the subscripts i , j , and t represent individual firms, the industry to which the firm belongs, and the year, respectively. Here, $IVlnword_{i,j,t}$ denotes the instrumental variable for the artificial intelligence usage status of firm i , $lnword_{i,j,t}$ represents the artificial intelligence usage status of firm i , and $lnwordGrowth_t$ indicates the national artificial intelligence growth rate in year t . A two-stage least squares (2SLS) estimation is employed to address potential endogeneity concerns, with the resulting coefficients summarized in Table 6. (See Table 7.)

The regression results show that the effect of the first-stage instrumental variable on firms' artificial intelligence usage is significantly positive. After incorporating the instrumental variable, firms' AI usage remains significantly positive at the 1 % level in terms of its impact on investment efficiency. The first-stage Cragg-Donald Wald F-statistic substantially exceeds the Stock-Yogo critical threshold, allowing us to reject the null hypothesis of weak instruments.

4.3. Categorized tests of corporate investment efficiency

Building upon the endogeneity tests, this paper further conducts categorized tests of corporate investment efficiency. Specifically, following Richardson's (2006) actual investment level model, the proxy variable for corporate investment efficiency is divided into overinvestment and underinvestment based on the positive or negative residual terms, and regression analyses are performed to validate the baseline regression results.

The regression results indicate that the firm's adoption of AI has a significantly positive impact on both overinvesting and underinvesting firms, demonstrating that AI exerts a notable effect on mitigating both overinvestment and underinvestment. The regression coefficients are 0.005 and 0.002, respectively, suggesting that AI's effect on curbing corporate overinvestment is stronger than its role in alleviating underinvestment. These categorized tests of investment efficiency further corroborate the baseline regression findings of this paper, namely that AI can enhance firm investment efficiency.

4.4. Robustness tests

To ensure the robustness of the baseline regression findings, this study conducts several robustness checks, including: (1) substituting the dependent variable with alternative measures; (2) using different proxies for the explanatory variable; (3) incorporating additional fixed effects; and (4) introducing further control variables.

4.4.1. Replacing the dependent variable

This paper adopts Tobin's Q as a proxy for firm growth, specifically replacing the revenue growth rate, to estimate the residual term of

Table 6
IV Results.

Variable	(1)	(2)
	First stage	Second stage
	<i>lnword</i>	<i>Inveff</i>
<i>IVlnword</i>	0.884*** (0.001)	
<i>lnword</i>		0.003*** (0.001)
<i>Control variable</i>	Yes	Yes
<i>Constant</i>	Yes	Yes
<i>Industry FE</i>	Yes	Yes
<i>Year FE</i>	Yes	Yes
<i>Observations</i>	24,768	24,768

This table shows the instrumental variable method test results for the impact of AI on firm investment efficiency, with the dependent variable in the second-stage regression being *Inveff*.

Table 7

Categorized Test Results of Corporate Investment Efficiency.

Variable	(1)	(2)
	Overinvestment	Underinvestment
	<i>Inveff</i>	<i>Inveff</i>
<i>lnword</i>	0.005*** (0.001)	0.002** (0.021)
<i>Control variable</i>	Yes	Yes
<i>Constant</i>	Yes	Yes
<i>Industry FE</i>	Yes	Yes
<i>Year FE</i>	Yes	Yes
<i>R²</i>	0.120	0.188
<i>Observations</i>	9361	15,407

This table shows the categorized test results of the effect of AI on corporate investment efficiency, with the explained variable being *Inveff*.

Richardson's (2006) actual investment level model. This residual term serves as a proxy for firm investment efficiency in the robustness test. The regression results obtained through this method are presented in Table 8.

The regression analysis reveals that the coefficient for *lnword* is 0.003, achieving statistical significance at the 1 % level. This outcome demonstrates that even after substituting the dependent variable, the positive effect of corporate AI adoption on investment efficiency persists, thereby reinforcing the robustness of the baseline regression findings.

4.4.2. Replacing the explanatory variable

Additionally, this paper matches the AI-related terms in Table 2 with firms' patents to measure the number of AI-related patents held by listed companies. The annual count of AI-related patents for each firm is incremented by one and then logarithmically transformed to serve as an alternative measure of firms' AI adoption, replacing the original explanatory variable in the regression. Table 9 presents the results.

The estimation results reveal that the coefficient on *lnpatent* is 0.002 and statistically significant at the 1 % level. This suggests that when AI-related patent counts are employed as a proxy for firms' AI adoption, the positive effect of AI adoption on investment efficiency persists. These findings provide additional evidence in support of the robustness of the baseline regression results.

4.4.3. Supplementing additional fixed effects

Additionally, the study incorporates industry-by-time and region-by-time fixed effects separately as part of robustness checks. The results indicate that even after controlling for these more granular fixed effects, the positive relationship between corporate AI adoption and investment efficiency remains statistically significant, further confirming the reliability of the baseline findings.

Table 8

Results of the robustness test with alternative explained variables.

Variable	<i>Inveff</i>
<i>lnword</i>	0.003*** (0.001)
<i>Control variable</i>	Yes
<i>Constant</i>	Yes
<i>Industry FE</i>	Yes
<i>Year FE</i>	Yes
<i>R²</i>	0.068
<i>Observations</i>	24,768

This table presents the robustness test results of AI's impact on corporate investment efficiency using alternative explained variables, where the explained variable is *Inveff*.

Table 9
Robustness Test Results of Replacing Explanatory Variables.

Variable	<i>Inveff</i>
<i>Inpatent</i>	0.002*** (0.001)
<i>Control variable</i>	Yes
<i>Constant</i>	Yes
<i>Industry FE</i>	Yes
<i>Year FE</i>	Yes
<i>R²</i>	0.123
<i>Observations</i>	17,460

This table presents the robustness test results of replacing explanatory variables for examining the impact of AI on firm investment efficiency, where the explanatory variable is *Inpatent*.

4.4.4. Adding control variables

Finally, this paper introduces additional control variables at the regional level, including regional GDP per capita and regional marketization level. Regional GDP per capita is measured by taking the logarithm of provincial GDP per capita. The regional marketization level is sourced from the China Marketization Index Database, with data extrapolated to 2023 based on historical figures for robustness testing.

The regression results³ show that after adding regional-level control variables, the coefficient on *Inword* is 0.003 and statistically significant at the 1 % level. This indicates that, even after controlling for relevant covariates, the positive effect of firms' AI adoption on investment efficiency persists, offering further confirmation of the robustness of the baseline regression results.

5. Heterogeneity analysis

5.1. Knowledge density heterogeneity

This study further explores the heterogeneity in the impact of AI on corporate investment efficiency by examining differences across firms with varying levels of knowledge intensity, R&D intensity, and internal control quality. To begin, companies are categorized into high and low knowledge density groups based on whether the proportion of employees holding at least a bachelor's degree in a given year is above or below the median. A binary variable is created each year to indicate whether a firm is above or below this median. These annual binary indicators are then aggregated to generate a new variable, from which a new median is calculated. Based on this, firms are reclassified into high and low knowledge density groups for heterogeneity analysis. Table 10 shows these results. The results indicate that the impact of AI adoption on firm investment efficiency is more pronounced among high knowledge density firms. This is likely because, compared to their low knowledge density counterparts, high knowledge density firms are better equipped to integrate AI tools efficiently and optimize workflows, thereby enhancing human-machine collaboration. As a result, these firms are more capable of mitigating inefficiencies in investment decisions.

5.2. R&D density heterogeneity

Firms are classified into high and low R&D density groups based on the median proportion of R&D and technical personnel in a given year. A binary variable is created annually to indicate whether a firm is above or below this median. These annual binary indicators are then summed to generate a new variable, and firms are reclassified into high and low R&D density groups based on the median of this aggregated variable. As

shown in Table 10, the impact of AI adoption on corporate investment efficiency is more significant among high R&D density firms. This is because firms with higher R&D density tend to have stronger capabilities in technology development and application, enabling them to effectively deploy, optimize, and deeply integrate AI systems into their core business processes. As a result, the marginal improvement in investment efficiency is greater for these firms compared to those with lower R&D density.

5.3. Internal control heterogeneity

Firms are classified into two groups—those with and without internal control deficiencies—for heterogeneity analysis. The findings in Table 10 indicate that the impact of AI adoption on investment efficiency is more significant among firms without internal control deficiencies. A sound internal control system provides a strong foundation for AI application by ensuring high-quality data, standardized processes, and an effective risk management framework, all of which contribute to improvements in investment efficiency. In contrast, firms with internal control deficiencies often suffer from poor data quality, disorganized processes, and weak oversight, which significantly hinder the effective implementation and performance of AI technologies.

6. Mechanism identification test

The regression results from the previous section indicate that AI has a significant positive effect on firm investment efficiency, passing endogeneity and robustness tests. Moreover, this positive effect exhibits heterogeneity across factors such as the region where the enterprise is located, the type of enterprise, and the scale of the enterprise. Building upon the earlier research, this section further analyzes the mechanism through which AI influences corporate investment efficiency—specifically, the pathways by which AI enhances corporate investment efficiency. This section employs a mediation effect model, using corporate innovation level and corporate financial condition as mediating variables to conduct mechanism identification tests.

6.1. Firm innovation level

Firm innovation is quantified as the natural logarithm of one plus the annual count of valid patents granted to each firm. The corresponding regression outcomes are shown in Table 11. The regression results provide evidence of a mediating mechanism, indicating that AI enhances corporate investment efficiency partly through fostering innovation.

6.2. Corporate financial condition

This study employs the distance to default (DD) as a proxy for corporate financial condition. Specifically, a higher DD indicates better financial health and a lower probability of default, while a lower DD suggests poorer financial condition and a higher likelihood of default. The DD-related data used in this section are sourced from the CSMAR database. The DD is calculated through iterative and simultaneous equation methods in Matlab, with reference to the KMV model's definition of DD. Table 12 shows the results. The regression results reveal that AI can enhance firm investment efficiency by improving financial conditions.

7. Further examination of economic consequences

To further investigate the impact of improved corporate investment efficiency on value creation capabilities, as well as whether artificial intelligence contributes to corporate value creation, we use *EVA* as a proxy indicator for our study. *EVA* denotes the value creation capability of the firm, measured by the firm's economic value added. The economic value added of a firm is calculated as the net operating profit after taxes

³ Due to space limitations, the results are not included in the main text.

Table 10
Results of Heterogeneity Analysis.

Variable	(1)	(2)	(3)	(4)	(5)	(6)
	High knowledge-intensity firms	Low knowledge-intensity firms	High R&D-intensity firms	Low R&D-intensity firms	Firms with sound internal controls	Firms with internal control deficiencies
	<i>Inveff</i>	<i>Inveff</i>	<i>Inveff</i>	<i>Inveff</i>	<i>Inveff</i>	<i>Inveff</i>
<i>lnword</i>	0.003*** (0.001)	0.002* (0.061)	0.003*** (0.002)	0.001 (0.321)	0.003*** (0.002)	0.001* (0.073)
<i>Control variable</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>Constant</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>Industry FE</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>Year FE</i>	Yes	Yes	Yes	Yes	Yes	Yes
<i>R</i> ²	0.136	0.101	0.119	0.125	0.117	0.133
<i>Observations</i>	12,878	9202	12,418	9662	13,598	6749

This table presents the results of heterogeneity analysis on the influence of AI on corporate investment efficiency, with *Inveff* as the explained variable.

Table 11
Mechanism Test 1.

Variable	(1)	(2)
	<i>Inveff</i>	<i>Inveff</i>
<i>lnword</i>	0.003*** (0.001)	0.002* (0.082)
<i>Innovation capability</i>		0.001*** (0.001)
<i>Control variable</i>	Yes	Yes
<i>Constant</i>	Yes	Yes
<i>Industry FE</i>	Yes	Yes
<i>Year FE</i>	Yes	Yes
<i>R</i> ²	0.113	0.094
<i>Observations</i>	23,070	8114

This table presents the test results of the corporate innovation level mechanism regarding the impact of AI on firm investment efficiency.

Table 12
Mechanism Test 2.

Variable	(1)	(2)
	<i>Inveff</i>	<i>Inveff</i>
<i>lnword</i>	0.003*** (0.001)	0.001* (0.089)
<i>Financial health</i>		0.001* (0.072)
<i>Control variable</i>	Yes	Yes
<i>Constant</i>	Yes	Yes
<i>Industry FE</i>	Yes	Yes
<i>Year FE</i>	Yes	Yes
<i>R</i> ²	0.116	0.117
<i>Observations</i>	21,898	21,898

This table presents the test results of corporate financial condition mechanisms regarding the impact of AI on firm investment efficiency.

minus the product of the firm's total capital and the weighted average cost of capital. The regression results using *EVA* as the explained variable are shown in Table 13.

The regression results indicate that the regression coefficients of both corporate investment efficiency and artificial intelligence usage on firm value creation are positive at the 1 % significance level. In terms of economic significance, holding other conditions constant, a one-unit increase in corporate investment efficiency leads to an average 4.2 % increase in economic value. Similarly, a one-unit improvement in artificial intelligence usage results in an average 0.2 % increase in economic value. This demonstrates that both the application of artificial intelligence and the enhancement of investment efficiency contribute to the improvement of firm value creation. It also substantiates the importance of corporate intelligent transformation—specifically, the adoption and promotion of artificial intelligence within firms—as well as the

Table 13
Further Economic Consequence.

Variable	(1)	(2)
	<i>EVA</i>	<i>EVA</i>
<i>Inveff</i>	0.042*** (0.002)	
<i>lnword</i>		0.002*** (0.003)
<i>Control variable</i>	Yes	Yes
<i>Constant</i>	Yes	Yes
<i>Industry FE</i>	Yes	Yes
<i>Year FE</i>	Yes	Yes
<i>R</i> ²	0.588	0.588
<i>Observations</i>	23,952	23,952

This table presents the regression results of the study on the impact of firm investment efficiency and AI on corporate value creation capability, with *EVA* as the explained variable.

enhancement of investment efficiency in the process of value creation.

8. Conclusion

The findings of this study reveal that artificial intelligence (AI) can significantly enhance corporate investment efficiency, and the results have passed endogeneity and robustness tests. The heterogeneity analysis reveals that the beneficial impact of AI on investment efficiency is stronger among firms with high knowledge intensity, elevated R&D investment, and robust internal control systems. Moreover, AI enhances corporate investment efficiency by boosting innovation capacity and improving financial health. Lastly, tests of economic consequences confirm that both increased investment efficiency and AI adoption play important roles in driving corporate value creation.

This study offers several actionable insights for policymakers seeking to enhance corporate investment efficiency through the adoption of AI. First, governments should provide clear strategic guidance and direct financial support to facilitate firms' digital and intelligent transformation. This could be achieved through the creation of a dedicated AI Transformation Fund or a voucher-based subsidy system to help offset the initial costs of AI adoption, particularly for small and medium-sized enterprises (SMEs) and firms in traditional manufacturing sectors. By lowering financial barriers to entry, such initiatives directly address the resource constraints identified in our analysis. Second, policy support should be prioritized and targeted. Resources from the proposed fund should be directed toward firms with low knowledge intensity, limited R&D spending, or weaker internal control systems—characteristics that, as our findings show, limit the natural benefits of AI adoption. Importantly, financial assistance to these firms should be contingent on parallel efforts to upskill their workforce or strengthen internal governance structures, thus fostering the complementary capabilities necessary for

AI to be effective. Third, to mitigate challenges related to knowledge-intensity heterogeneity, the government should collaborate with industry and academic institutions to establish a national AI Skills Certification and Training System. This system would focus on reskilling and upskilling the existing workforce, cultivating a broader talent pool capable of working effectively alongside AI systems. Such efforts are crucial for enabling more firms to integrate AI tools successfully and optimize human-machine collaboration.

This study also opens several avenues for future research. First, the measure of AI adoption could be refined by distinguishing between different types of AI applications (e.g., machine learning vs. robotics) to examine their varying impacts on investment and other corporate outcomes. Second, future studies could investigate how AI influences labor composition within firms and how this reorganization, in turn, affects decision-making efficiency. Third, as more data becomes available, research could explore the long-term and general equilibrium effects of widespread AI adoption on industry dynamics and market competition. Finally, examining the relationship between AI governance, ethical guidelines, and corporate financial decisions presents another important area for future study.

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Data availability

Data will be made available on request.

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